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THE ELABORATION LIKELIHOOD MODEL OF PERSUASION

Richard E. Petty

DEPARTMENT OF PSYCHOLOGY
UNIVERSITY OF MISSOURI
COLUMBIA, MISSOURI

John T. Cacioppo

DEPARTMENT OF PSYCHOLOGY
UNIVERSITY OF IOWA
IOWA CITY, IOWA

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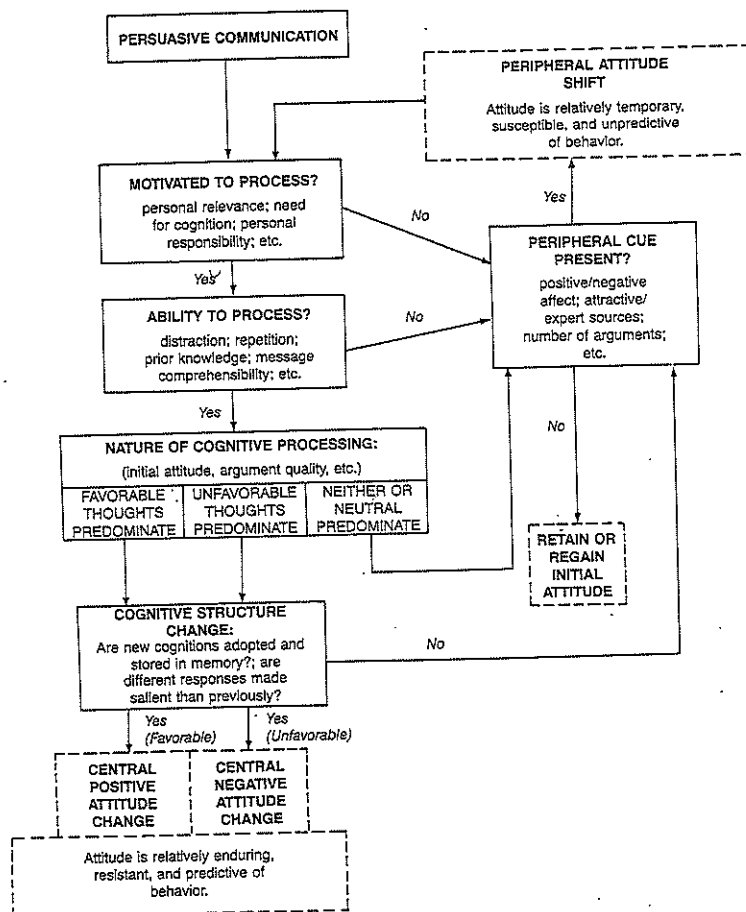


Fig. 1. Central and peripheral routes to persuasion. This figure depicts the two anchoring endpoints on the elaboration likelihood continuum (adapted from Petty, 1977; Petty & Cacioppo, 1978, 1981a).

the ELM, which is based on these two routes (Petty & Cacioppo, 1981a). In addition, we have addressed the various applications of this model to fields such as psychotherapy and counseling (Cacioppo, Petty, & Stoltenberg, 1985; Petty, Cacioppo, & Heesacker, 1984) and mass media advertising and selling (Cacioppo,

po, & Petty, 1985; Petty & Cacioppo, 1983a, 1984b; Petty, Cacioppo, & Schumann, 1984). In the remainder of this article we will outline the ELM as a series of postulates that make explicit the guiding assumptions and principles of the model. We will also provide a methodology for testing the major processes outlined by the ELM and we will review research which provides evidence relevant to the framework.

Before outlining our model of attitude change, however, it is important to define our use of the term *attitude*. Consistent with the positions of other attitude theorists (e.g., Thurstone, 1928), we regard attitudes as general evaluations people hold in regard to themselves, other people, objects, and issues. These general evaluations can be based on a variety of behavioral, affective, and cognitive experiences, and are capable of influencing or guiding behavioral, affective, and cognitive processes. Thus, a person may come to like a new political candidate because she just donated \$100 to the campaign (behavior-initiated change), because the theme music in a recently heard commercial induced a general pleasantness (affect-initiated change), or because the person was impressed with the candidate's issue positions (cognitive initiated change). Similarly, if a person already likes a political candidate he may agree to donate money to the campaign (behavioral influence), may feel happiness upon meeting the candidate (affective influence), and may selectively encode the candidate's issue positions (cognitive influence).

II. Postulate 1: Seeking Correctness

Our first postulate and an important guiding principle in the ELM agrees with Festinger's (1950) statement that:

People are motivated to hold correct attitudes.

Incorrect attitudes are generally maladaptive and can have deleterious behavioral, affective, and cognitive consequences. If a person believes that certain objects, people, or issues are “good” when they are in fact “bad,” a number of incorrect behavioral decisions and subsequent disappointments may follow. As Festinger (1954) noted, the implication of such a drive is that “we would expect to observe behavior on the part of persons which enables them to ascertain whether or not their opinions are correct” (p. 118). In his influential theory of social comparison processes, Festinger (1954) focused on how people evaluated the correctness of their opinions by comparing them to the opinions of others. In Section IX,B we address how the ELM accounts for attitude changes induced by exposure to the opinions of varying numbers of other people. But first we need to outline our other postulates.

If no strong affective cues are presented, it is still possible for people to form a "reasonable" attitude without relying on scrutiny of the issue-relevant arguments presented by relying on various persuasion rules or inferences that may be either rather simple or relatively complex. For example, according to self-perception theory (Bem, 1972), people may come to like or dislike an object as a result of a simple inference based on their own behavior (e.g., if I bought it, I must like it). According to the heuristic model of persuasion (Chaiken, 1980; Eagly & Chaiken, 1984), people may evaluate messages by employing various rules that they have learned on the basis of past experience (e.g., people agree with people they like). Social judgment theory (Sherif & Sherif, 1967) proposes that people evaluate messages mostly on the basis of their perceived position—messages are contrasted and rejected if they appear too discrepant (fall in the latitude of rejection), but are assimilated and accepted if they appear closer to one's initial position (fall in the latitude of acceptance; Pallak, Mueller, Dollar, & Pallak, 1972).

In addition to the relatively simple acceptance/rejection rules proposed by the preceding models, attitude change may be affected by more complex reasoning processes, such as those based on balance theory (Heider, 1946; Insko, 1984) or certain attributional principles (e.g., Kelley, 1967; Eagly, Wood, & Chaiken, 1978). Importantly, even reliance on more complex inferences obviates the need for careful scrutiny of the issue-relevant arguments in a message. In other words, each of these processes (e.g., self-perception, assimilation, balance) is postulated to be sufficient to account for attitude change without requiring a personal evaluation of the issue-relevant arguments.² In sum, we have proposed that when either motivation or ability to process issue-relevant arguments is low, attitudes may be changed by associating an issue position with various affective cues, or people may attempt to form a reasonable opinion position by making an inference about the likely correctness or desirability of a particular attitude position based on cues such as message discrepancy, one's own behavior, and the characteristics of the message source.

B. DEVELOPMENTAL TRENDS IN ELABORATION

Interestingly, the attitude change processes that we have just described form an elaboration continuum which likely coincides with the manner in which attitude change processes develop through adulthood. Specifically, the very young child probably has relatively little motivation to think about the true merits of people, objects, and issues, and even less ability to do so. Thus, attitudes may

²Insko (1981) extended balance theory to include a person's consideration of issue-relevant arguments. This more general balance formulation therefore broadens the theory beyond peripheral processing.

be affected primarily by what feels good or bad. As children mature, they become more motivated to express correct opinions on certain issues, but their ability to scrutinize issue-relevant arguments may still be poor due to lack of knowledge. Therefore, they may be particularly reliant on certain cognitive rules based on personal experience such as, "My mother knows what's right," or "If I play with it, I must like it." Consistent with this reasoning, children have been shown to be more susceptible to appeals based on behavioral cues and self-perceptions than issue-relevant argumentation (e.g., Miller, Brickman, & Bolen, 1975).

Finally, as people move into adulthood, interests become more focused and the consequences of holding correct opinions on certain issues increase. In addition, as people's acquired knowledge and cognitive skills grow, this renders them more able to critically analyze issue-relevant information on certain topics and makes them less reliant than children on certain primitive heuristics (cf. Ross, 1981). As we noted earlier, of course, although people may have the requisite ability and motivation to scrutinize certain attitude issues, they will lack motivation and ability on others. Thus, simple inferences and affective cues may still produce attitude change in adults.

In sum, one's initial evaluations are likely to be largely hedonistic since, lacking the motivation and/or ability to consider issue-relevant arguments, attitudes will be based primarily on positive and negative affective cues associated with the attitude object. As development proceeds, some attitudes may be formed on the basis of simple inferences, decision rules, and social attachments. Finally, the formation and change of some attitudes become very thoughtful processes in which issue-relevant information is carefully scrutinized and evaluated in terms of existing knowledge. Importantly, our sequence of the developmental stages of influence is consistent with other developmental models of judgment. For example, in discussing the development of moral standards, Kohlberg (1963) identifies three developmental levels. At the first level (preconventional), moral evaluations are based primarily on the affective consequences of an act. At level 2 (conventional), evaluations of acts are based primarily on socially accepted rules and laws. Finally, at level 3 (postconventional), an evaluation of an act is based on a person's idiosyncratic but well-articulated moral code. The parallels to our stages of influence are obvious.

Although we have argued that there is a continuum of message elaboration ranging from none to complete, and that different attitude change processes may operate along the continuum, it is also important to note that these different theoretical processes can be viewed as specifying just two qualitatively distinct routes to persuasion. The first route, which we have called the "central route," occurs when motivation and ability to scrutinize issue-relevant arguments are relatively high. The second, or "peripheral route," occurs when motivation and/or ability are relatively low and attitudes are determined by positive or negative cues in the persuasion context which either become directly associated

Once the messages meet the criterion of eliciting the appropriate profile of thoughts, they are checked for other characteristics. First, a panel of subjects rates the messages for overall believability. Our goal is to develop arguments that are strong and weak, but that do not strain credulity. (This is not to say that our arguments are necessarily veridical—just reasonably plausible to our subjects.) Next, people from the relevant subject pool rate the messages for comprehensibility, complexity, and familiarity. Again, our goal is to develop strong and weak messages that are roughly equivalent in their novelty and in our subjects' ability to understand them. The top panel of Fig. 2 depicts the results of a hypothetical study in which some extramessage "treatment" has no effect on persuasion. In this study, only the quality of the message arguments determined the extent of attitude change. We will compare this simple result with the other possibilities depicted in Fig. 2 in the remainder of this article.

B. PERIPHERAL CUES

According to the Elaboration Likelihood Model, one way to influence attitudes is by varying the quality of the arguments in a persuasive message. Another possibility, however, is that a simple cue in the persuasion context affects attitudes in the absence of argument processing. As we noted earlier, some cues will do this because they trigger relatively primitive affective states that become associated with the attitude object. Various reinforcing (e.g., food; Janis, Kaye, & Kirschner, 1965) and punishing (e.g., electric shock; Zanna, Kiesler, & Pilkonis, 1970) stimuli have proved effective in this regard. Other cues work, however, because they invoke guiding rules (e.g., balance; Heider, 1946) or inferences (e.g., self-perception; Bem, 1972).

Since cues are postulated to affect attitude change without affecting argument processing, it is possible to test manipulations as potential cues by presenting them to subjects with the advocated position only (i.e., without accompanying persuasive arguments), as in prestige suggestion (see Asch, 1948). If the manipulation is a potential cue, it should have the ability to affect attitudes in the absence of any arguments. Alternatively, one could present an incomprehensible message (e.g., in a foreign language) on some topic along with the potential cue (e.g., speed of speech; Miller *et al.*, 1976). Subjects could be asked to rate, for example, how likely it is that the speaker is convincing. Again, if the cue is operative, it should be capable of affecting judgments even if there are no arguments to process. Finally, a simple procedure might involve merely describing various potential cues to subjects (e.g., a message with 1 vs. 10 arguments; a message from an attractive vs. an unattractive source) and asking them which would more likely be acceptable and/or persuasive. These procedures would not, of course, indicate *why* a cue was effective (e.g., were the judgments due to affective association or the invocation of a simple decision rule?), nor would they

eliminate the possibility that more thoughtful processes were involved (e.g., subjects might attempt to generate arguments consistent with the position; cf., Burnstein, Vinokur, & Trope, 1973). However, these procedures would indicate whether or not a manipulation has the *potential* to serve as a peripheral cue.

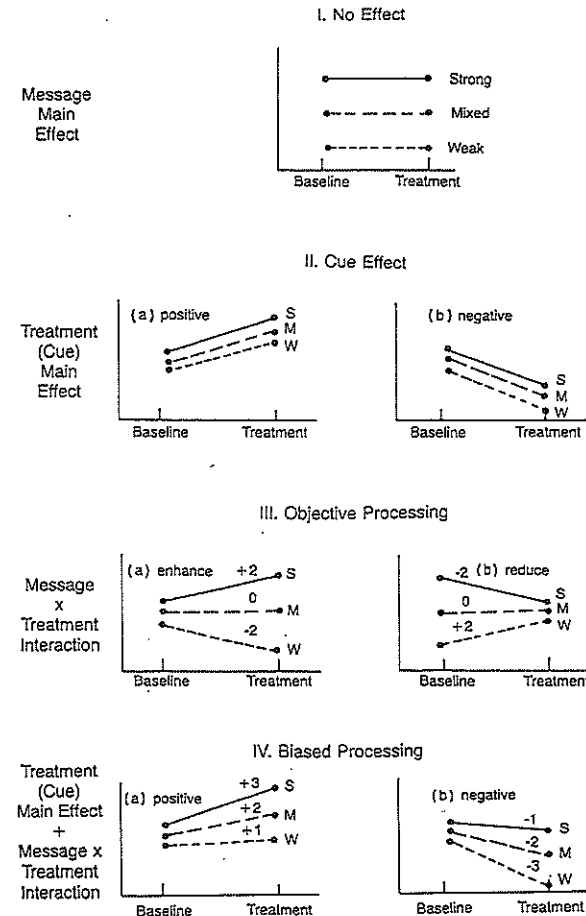


Fig. 2. Impact of variables on attitude change according to the ELM. Under conditions of high elaboration likelihood, attitudes are affected mostly by argument quality (I). Under conditions of low elaboration likelihood, attitudes are affected mostly by peripheral cues (II). Under conditions of moderate elaboration likelihood, variables may enhance or reduce message processing in either a relatively objective (III) or relatively biased (IV) manner (adapted from Petty & Cacioppo, 1984c).

Affecting motivation and/or ability to process a message in a relatively objective manner can do so by either enhancing or reducing argument scrutiny.

As we hinted above, our empirical method of defining argument quality allows us to assess the extent to which a variable affects argument processing and the extent to which this processing is relatively objective or biased. We shall consider first the expected consequences of variables affecting relatively objective processing.

Assume for the moment that we have created a control condition in which motivation or ability to process issue-relevant arguments is rather low. Subjects should show relatively little differentiation of strong from weak arguments in this condition. However, if a manipulation enhances argument processing in a relatively objective manner, then subjects should show greater differentiation of strong from weak arguments. More specifically, a message with strong arguments should produce more agreement when it is scrutinized carefully than when scrutiny is low, but a message with weak arguments should produce less overall agreement when scrutiny is high rather than low. This pattern of results is depicted in the left half of Panel III in Fig. 2. In a similar fashion, we can assess the extent to which a variable disrupts processing in a relatively objective manner. Consider a situation in which subjects are processing the message arguments quite diligently. These subjects should show considerable differentiation of strong from weak arguments. However, if argument processing is disrupted, due either to reduced motivation or ability, argument quality should be a less important determinant of persuasion. More specifically, a strong message should induce less agreement when processing is disrupted than when it is not, but a weak message should produce more agreement when processing is disrupted than when it is not. The right half of Panel III in Fig. 2 depicts this pattern. In addition to subjects' attitudes being more differentiated to weak and strong messages when argument processing is high rather than low, the profile of subjects' thoughts also should show greater differentiation of arguments when processing is high rather than low.

In sum, by manipulating argument quality along with some other variable, it is possible to tell whether that variable enhances or reduces argument processing in a relatively objective manner. If the variable enhances argument processing, subjects' thoughts and attitudes should be more polarized when the variable is present rather than absent, but if the variable reduces argument processing, subjects' thoughts and attitudes should be less polarized when the variable is present rather than absent. Before moving on to our postulates concerning peripheral cues and biased processing, we review some evidence that variables can affect persuasion by affecting the extent of argument processing in a relatively objective manner.

A. DISTRACTION

Research on distraction's effect on persuasion can be traced to an intriguing study by Allyn and Festinger (1961), in which high school students were presented with a speech which argued that teenage drivers are dangerous. The students were either forewarned of the opinion topic and told that their opinions would be assessed (opinion orientation) or were told simply that they were to assess the personality of the speaker (personality orientation). Although these two conditions did not differ in the average opinion change they induced, when analyses were conducted on the most involved subjects (those with extreme opinions or those who said the issue was important), a significant difference was found such that there was more persuasion in the personality than in the opinion orientation condition. Two possible explanations for this effect were offered. The initial explanation favored by Allyn and Festinger was that the "forewarning" in the opinion orientation condition stimulated the involved students to counterargue and/or derogate the source (see also Freedman & Sears, 1965). A second explanation, proposed initially by Festinger and Maccoby (1964), was that the involved subjects in the personality orientation condition were distracted from the counterarguing and/or source derogating in which they normally would have engaged.

In the years since the Allyn and Festinger experiment, a considerable number of studies have accumulated on both "forewarning" and "distraction," and it is now clear that both effects are viable. In this section we apply the ELM framework to "distraction" and discuss how this variable works by affecting information processing in a relatively *objective* manner. In section VII,B we apply the ELM to "forewarning" and address how this variable also works by affecting information processing, but in a relatively *biased* manner.

In 1973, Baron, Baron, & Miller reviewed the accumulated research on "distraction" and concluded that although many individual studies were susceptible to a wide variety of mediational interpretations, there were just two theoretical explanations that could account for the existing data parsimoniously. One explanation was the disruption of counterarguing interpretation favored by Festinger and Maccoby. Another interpretation offered by Baron *et al.*, however, was based, ironically, on Festinger's (1957) theory of cognitive dissonance. Baron *et al.* argued that distraction manipulations require subjects to exert more effort than usual in order to understand the message. Furthermore, "since choosing to hear a counterattitudinal message can be viewed as attitude-discrepant behavior, the effort required to comprehend a counterattitudinal message will directly determine the amount of dissonance created by the choice" (p. 317). One way for subjects to reduce this dissonance, of course, is for them to justify their effort by overvaluing the communication.

objective
processing
strong > weak

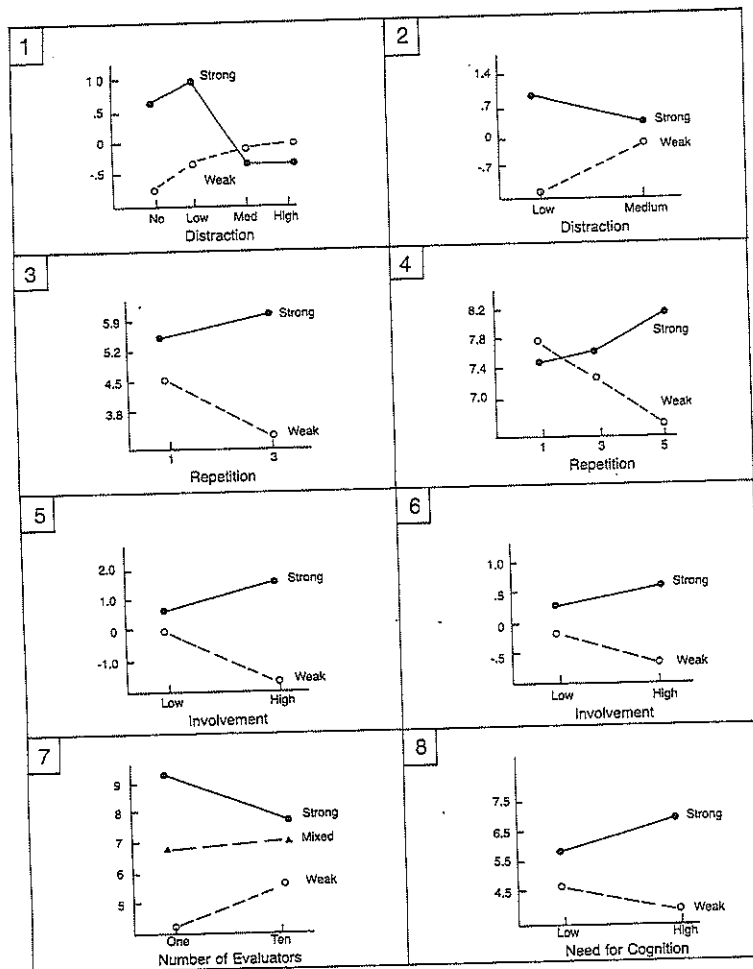


Fig. 3. Variables that may enhance or reduce elaboration in a relatively objective manner. (1) Effects of distraction on attitudes following strong and weak counterattitudinal messages (data from Petty, Wells, & Brock, 1976; Experiment 1). (2) Effects of distraction on attitudes following strong and weak proattitudinal messages (data from Petty, Wells, & Brock, 1976; Experiment 2). (3) Effects of message repetition on initial attitudes following strong and weak messages (data from Cacioppo & Petty, 1985). (4) Effects of message repetition on delayed attitudes following strong and weak messages (data from Cacioppo & Petty, 1980a, Experiment 2). (5) Effects of personal relevance on

B. REPETITION

Repetition of stimuli has been shown to increase liking (e.g., Zajonc, 1968), decrease liking (e.g., Cantor, 1968) and have no effect on attitudes (e.g., Belch, 1982). The most common finding in the persuasion literature, however, is that repeating a persuasive communication tends to first increase and then decrease agreement (e.g., Cacioppo & Petty, 1979b; Calder & Sternthal, 1980; Gorn & Goldberg, 1980). A variety of theoretical accounts has been proposed for the effects of repeated exposure, including message learning, response competition, and others (see reviews by Harrison, 1977; Sawyer, 1981).

Based on the accumulated research, we proposed that message repetition guides a sequence of psychological reactions to a persuasive communication best conceptualized as a two-stage attitude-modification process (Cacioppo & Petty, 1979b). In the first stage, repeated presentations of a message provide recipients with a greater opportunity to consider the implications of the content of the message in a relatively objective manner. Thus, just as distraction can disrupt information processing, repetition can enhance a person's ability to process the message arguments. The benefit of repetition should be most apparent when additional opportunities are needed to process a message, such as when ability to process the full implications of the message with only one exposure is low (e.g., the message is complex), or when motivation to process with one exposure is low. Once a person has considered the implications of the message, however, the second stage of information processing commences. In this second stage, the relatively objective processing of the first stage ceases as tedium and/or reactance are elicited by the excessive exposures. Both tedium and reactance will tend to result in decreased message acceptance either by serving as simple negative affective cues or by biasing the nature of information processing in a negative direction (see Section VII.C). In this section we explore the consequences of the first (objective) stage of information processing.

In order to provide a test of our view that moderate repetition can affect persuasion by increasing the opportunity to scrutinize arguments in a relatively objective manner, we conducted a study in which students were exposed to a message advocating that seniors at their university be required to take a comprehensive exam in their major area as a requirement for graduation (see Cacioppo

attitudes following pro- (strong) and counterattitudinal (weak) messages (data from Petty & Cacioppo, 1979b; Experiment 1). (6) Effects of personal relevance on attitudes following strong and weak counterattitudinal messages (data from Petty & Cacioppo, 1979b; Experiment 2). (7) Effects of personal responsibility on attitudes following strong, weak, and mixed messages (data from Petty, Harkins, & Williams, 1980; Experiment 2). (8) Effects of need for cognition on attitudes following strong and weak messages (data from Cacioppo, Petty, & Morris, 1983; Experiment 2).

than they really were and therefore more objectionable), but proattitudinal messages were *assimilated* (seen as closer to one's own position and therefore more acceptable).

Importantly, explanations of involvement based on social judgment theory did not consider the nature of the issue-relevant arguments presented in the communication. Instead, as involvement increased, a message was thought to induce increased assimilation (and acceptance) or increased contrast (and rejection) based on the particular position that it was judged to espouse. The ELM suggests an alternative analysis of the effects of personal involvement or relevance (Petty & Cacioppo, 1979b). Specifically, we suggested that as personal relevance increases, people become more motivated to process the issue-relevant arguments presented. As the personal consequences of an advocacy increase, it becomes more important for people to form a veridical opinion because the consequences of being incorrect are greater. Because of the greater personal implications people should be more motivated to engage in the cognitive work necessary to evaluate the true merits of the proposal.

Much of the early work on issue involvement was conducted by finding existing groups that differed in the extent to which an issue was important (as assessed by membership in issue-relevant groups), and thus was correlational in nature (e.g., Hovland, Harvey, & Sherif, 1957). More recent investigators have chosen to study issue relevance by varying the issue and message between subjects (e.g., Lastovicka & Gardner, 1979). For example, some undergraduate students would receive a message on a highly involving issue (e.g., increasing tuition), whereas others would receive a message on an issue of low relevance (e.g., increasing park acreage in a distant city; Rhine & Severance, 1970). Although this research is interesting in that these involvement classifications probably capture the personal relevance concept as it often occurs in the "real world," several interpretive problems are introduced. Specifically, distinctions based on different kinds of people or different issues may confound personal relevance with other factors (see discussion by Kiesler *et al.*, 1969). One particularly likely confound is that people in the high relevance groups or who receive the high relevance issues may be more familiar with the issue and may have more topic-relevant knowledge. Thus, in addition to possessing greater motivation to process the messages, it is likely that these subjects also have greater ability to do so. Thus, when a message contains information that is inconsistent with subjects' initial opinions, high relevance subjects should be more motivated and generally more able to generate counterarguments to the arguments presented. However, when a message contains information that is consistent with the subjects' initial attitudes, high relevance subjects should be more motivated and generally more able to elaborate the strengths of the arguments. In sum, it is possible that differences in message-relevant elaboration between high and low relevance subjects (rather than assimilation/contrast effects) may account for the different

effects obtained for pro- and counterattitudinal issues in previous research on personal involvement.

In order to test our formulation, we first sought to replicate previous research using a manipulation of personal relevance that did not include differences in familiarity with the issue and arguments as a component. Employing a procedure introduced by Apsler and Sears (1968), we had subjects in both high and low relevance groups receive the same message on the same topic, but high involvement subjects were led to believe that the advocacy would affect them personally, whereas low involvement subjects were led to believe that the advocacy would have no personally relevant implications.

In our initial experiment (Petty & Cacioppo, 1979b, Experiment 1), undergraduate students received either a proattitudinal message extolling the virtues of more lenient coed visitation hours on college campuses, or a counterattitudinal message contending that colleges should be more strict in their coed visitation policies. The message arguments were pretested so that the counterattitudinal message arguments were weak and elicited predominantly unfavorable thoughts, and the proattitudinal message arguments were strong and elicited predominantly favorable thoughts when subjects were instructed to think about them. To manipulate personal relevance, half of the subjects was told that the speaker was advocating that the change in visitation hours be implemented at their own university (Notre Dame), whereas the other half was told that the speaker advocated the change for a distant college (Juanita Junior College). As depicted in Box 5 of Fig. 3, a message direction/quality $\times$ relevance interaction was obtained on the measure of subjects' attitudes toward the change in visitation policy. When the message was counterattitudinal (and weak), increased relevance was associated with decreased acceptance, but when the message was proattitudinal (and strong), increased relevance was associated with greater acceptance.

Although this study provides evidence consistent with our view that increasing personal relevance enhances motivation to scrutinize message content, it is still possible that attitude change was mediated by assimilation/contrast effects since the strong arguments advocated a proattitudinal position and the weak arguments advocated a counterattitudinal one. To provide a stricter test of the Elaboration Likelihood Model, we conducted a second experiment (Petty & Cacioppo, 1979b, Experiment 2) in which all subjects were exposed to a counterattitudinal message advocating that college seniors should be required to pass a comprehensive exam in their major area as a requirement for graduation. For half of the subjects, the arguments in the message were strong and compelling, and for the other half, the arguments were weak and specious. Finally, for half of the subjects the speaker advocated that the exam policy be instituted at their own university (University of Missouri), and for half the speaker advocated implementation at a distant school (North Carolina State). The results were identical to

shared responsibility generated significantly fewer thoughts than those who were individually responsible. As might be expected if personal relevance motivates issue-relevant thinking (Petty & Cacioppo, 1979b), less loafing occurred in groups when the issue had high personal relevance.

The implications of this research for persuasion are straightforward: the greater the personal responsibility for evaluating an issue, the more people should be willing to exert the cognitive effort necessary to evaluate the issue-relevant arguments presented. To test this hypothesis, we asked undergraduates to provide peer feedback on editorial messages ostensibly written by journalism students (Petty, Harkins, & Williams, 1980, Experiment 2). Subjects were led to believe that they were either the only person responsible for evaluating an editorial or 1 of 10 people who shared the responsibility. Subjects received one of three versions of a message arguing that seniors should be required to pass a comprehensive exam in their major as a requirement for graduation. One message contained strong arguments, another contained weak arguments, and a third contained a mixture of arguments (and elicited a mixture of favorable and unfavorable thoughts). After reading the appropriate message, subjects provided an evaluation and listed their thoughts. The attitude results, graphed in Box 7 of Fig. 3, revealed a message quality $\times$ responsibility interaction. As personal responsibility for evaluation decreased, the quality of the arguments in the message became a less important determinant of the evaluations. More specifically, group evaluators were significantly more favorable toward the weak message, but were significantly less favorable toward the strong message than individual evaluators. As expected, evaluations of the mixed message were unaffected by the extent of responsibility.

E. NEED FOR COGNITION

Just as there are situational factors that influence the likelihood that individuals will think about and elaborate upon the arguments provided in a message, so too must there be individual factors governing message processing, and, indirectly, persuasion. Cohen, Stotland, and Wolfe (1955) introduced an individual difference called the "need for cognition," which they described as "a need to structure relevant situations in meaningful, integrated ways. It is a need to understand and make reasonable the experiential world" (p. 291). Early research on this construct suggested that people high in need for cognition made more discriminating judgments and were more motivated to think about persuasive communications (e.g., Cohen, 1957). Unfortunately, the objective tests used to gauge individual differences in need for cognition were never described in detail or published, and are apparently no longer available. Because of the great relevance of individual differences in motivation to think to the ELM and to

cognitive social psychology more generally, we developed and validated a new assessment instrument (Cacioppo & Petty, 1982; Cacioppo, Petty, & Kao, 1984). Specifically, in an initial study, we generated a pool of statements concerning a person's reactions to engaging in effortful thinking in a variety of situations (e.g., "I really enjoy a task that involves coming up with new solutions to problems") and tested them on two groups of people presumed to differ substantially in their tendencies to engage in and enjoy effortful cognitive endeavors (i.e., university faculty vs. assembly line workers). Thus, the need for cognition scale (NCS) was designed to distinguish individuals who dispositionally tend to engage in and enjoy effortful analytic activity from those who do not (see Cacioppo & Petty, 1982, 1984b, for further information about scale construction and validation).

The results of several studies indicate that individuals high in need for cognition do indeed enjoy relatively effortful cognitive tasks, even in the absence of feedback about performance. For example, in one study (Cacioppo & Petty, 1982; Experiment 4), subjects were given either simple or complex rules to use in performing a boring number circling task. Afterward, subjects were asked to express their attitudes about the task. Results revealed that subjects generally disliked the task, but a significant interaction revealed that individuals high in need for cognition tended to prefer the complex to the simple task whereas individuals low in need for cognition tended to prefer the simple to the complex task. In another study, subjects who were low in need for cognition "loafed" on a brainstorming task when they were part of a group that was responsible for generating uses for an object, but subjects who were high in need for cognition did not loaf on this cognitive task (i.e., they generated the same high number of uses whether they were solely or jointly responsible; Petty, Cacioppo, & Kasmer, 1985).

Again, the implications for responses to persuasive communications are straightforward. If people high in need for cognition tend to engage in and enjoy effortful cognitive activity, they should be particularly likely to evaluate a message by scrutinizing and elaborating the issue-relevant arguments presented. In order to test this hypothesis, we exposed high and low need for cognition subjects to a set of strong or weak arguments for a counterattitudinal position (raising tuition at their university; Cacioppo *et al.*, 1983, Experiment 2). After message exposure, subjects were asked to provide an overall evaluation of the message arguments and their personal opinion about the issue. Both measures indicated that subjects high in need for cognition scrutinized the message more carefully than subjects low in need for cognition. Specifically, the strong and weak messages induced more polarized evaluations and attitudes for high than low need for cognition subjects (attitude results are graphed in Box 8, Fig. 3). In addition, we reasoned that if subjects high in need for cognition were more likely to derive their attitudes through a considered evaluation of the arguments central

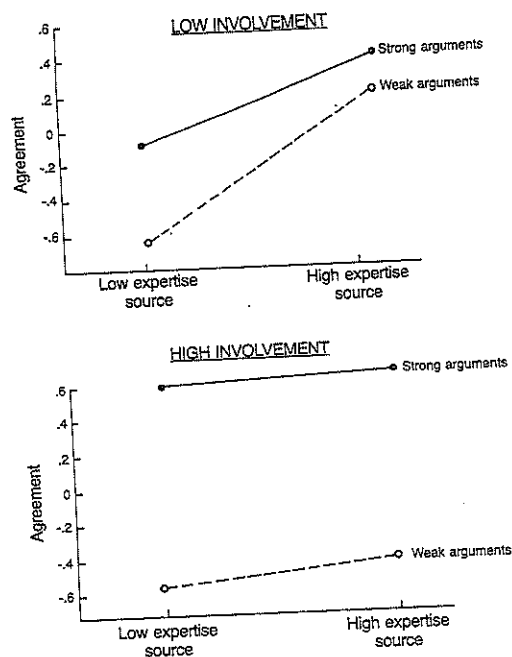


Fig. 4. Source factors under high and low relevance. (Left) Source expertise serves as a peripheral cue under low relevance conditions (top), but only argument quality affects attitudes under high relevance (bottom) (data from Petty, Cacioppo, & Goldman, 1981). (Right) Famous product endorsers serve as a peripheral cue under low relevance conditions (top), but only product quality information affects attitudes under high relevance (bottom) (data from Petty, Cacioppo, & Schumann, 1983).

method of message presentation. In this study (Petty, Cacioppo, & Schumann, 1983), undergraduates were asked to examine a booklet containing 12 magazine advertisements. Each of the ads was preceded by a brief description of the purpose of the ad. A variety of both familiar and unfamiliar ads appeared in the booklet, but the crucial ad was for a fictitious new product, "Edge disposable razors." Two things were done to either enhance or reduce the personal relevance of the ad for this product. In the high relevance groups, the ad was preceded by a description indicating that the product would be test marketed soon in the subjects' community. In the low relevance groups, the crucial ad was preceded by a description indicating that the product would be test marketed soon in several distant cities. In addition, all subjects were told before examining any

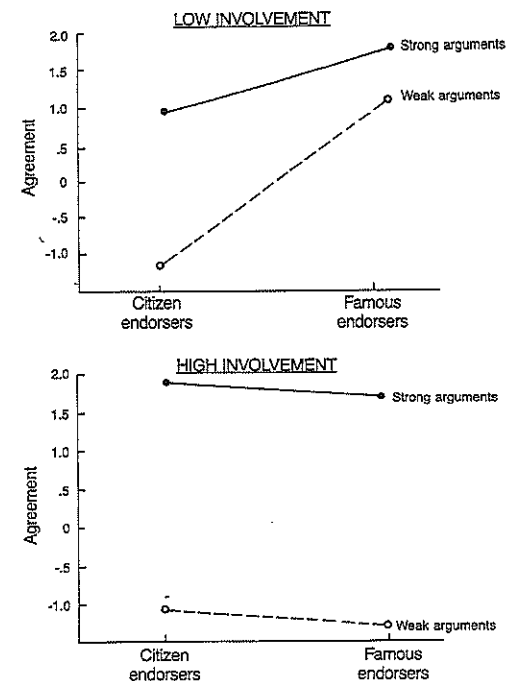


Fig. 4. (continued).

ads that at the end of the experiment they would be given a free gift for their participation. In the high relevance groups, they were told that they would be allowed to choose among several brands of disposable razors. In the low relevance groups, they were told that they would be selecting among brands of toothpaste (an ad for toothpaste appeared in the ad booklet). In sum, the high relevance subjects were not only led to believe that the crucial product would be available in their local area soon, but they also believed that they would make a decision about the product class. In contrast, the low relevance subjects believed that the product would not be available in their local area in the foreseeable future and did not expect to make a decision about that product class.

Four different versions of the razor ad were constructed. Two featured photographs of two well-known and -liked sports celebrities, and two featured middle-aged citizens described as Californians. The product endorsers served as the manipulation of the peripheral cue. Finally, two of the ads contained six

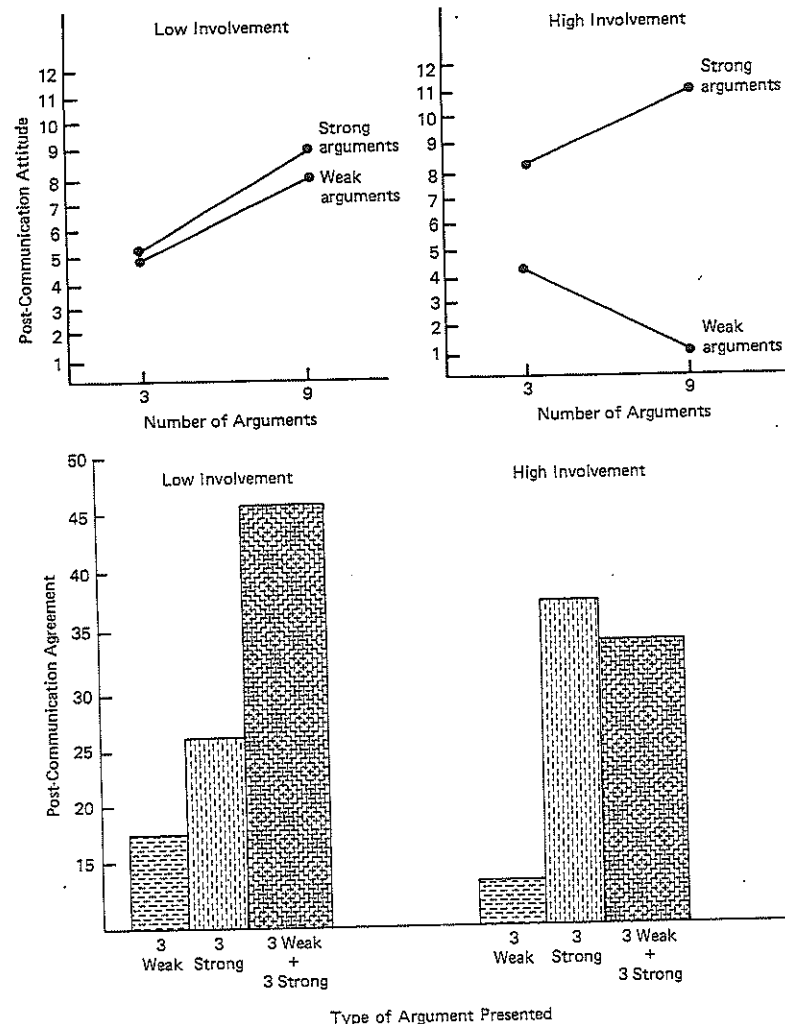


Fig. 5. Message factors under high and low relevance. (Top) Number of message arguments serves as a peripheral cue under low relevance conditions (left) but affects information processing under high relevance (right) (data from Petty & Cacioppo, 1984a; Experiment 2). (Bottom) Number of message arguments serves as a peripheral cue under low relevance (left) but affects information processing under high relevance (right) (data from Petty & Cacioppo, 1984a; Experiment 1).

cessing: when the arguments are strong increasing their number enhances persuasion, but when their quality is weak, increasing their number reduces persuasion.

In our second study, undergraduates were asked to read one of three messages. All of the messages concerned a faculty proposal to increase tuition, but in the high relevance conditions the proposal was for the students' own university, whereas in the low relevance conditions the proposal was for a distant but comparable university. The message that subjects read contained either three cogent arguments, three weak arguments or six arguments (three strong and three weak). After reading the assigned message, subjects indicated their attitudes toward the idea of raising tuition. Statistical comparison of the messages processed under high and low relevance conditions revealed the following (see bottom half of Fig. 5). When the issue was of low relevance, three strong arguments did not elicit more agreement than three weak arguments, but the message with six arguments (three strong and weak) elicited more agreement than either of the three-argument messages. When the message was highly relevant, however, three strong arguments did elicit more agreement than three weak arguments, but the six-argument message did not enhance persuasion over presenting three strong arguments. Again, argument quantity served as a cue under low relevance, but argument quality was more important under high relevance.

Langer, Blank, and Chanowitz (1978) explored the importance of the perception of arguments in a field study of compliance. All subjects in this study were standing in line to make copies when a confederate approached them with a request to make either 5 (low personal consequences) or 20 (high consequences) copies. The request was accompanied by either a valid reason ("I'm in a rush"), a "placebic" reason ("I have to make copies"), or no reason. Both kinds of reasons were more successful than no reason when the personal consequences were low (i.e., argument quality was unimportant), but the valid reason was significantly more potent than the placebic reason when the personal consequences were high. Folkes (1985) provided a partial replication of this effect. In two field studies using the inconsequential request (making five copies), respondents were equally willing to comply whether the request contained the valid or the placebic reason. In a third study, however, subjects were asked to guess how they would respond to the requests and to "think carefully before answering." When instructed to think before responding, the valid reason produced significantly more anticipation of compliance than the placebic reason.⁷ In sum,

⁷Although providing a partial replication of Langer *et al.* (1978), Folkes takes issue with Langer's assertion that the placebic information is processed "mindlessly." Folkes argues that if the reasons are processed automatically under low consequences conditions, then a poor reason should be as effective as a valid one. However, she found that a poor reason (e.g., "because I don't want to wait") was significantly less effective than a valid or placebic one under low consequences conditions. The ELM would predict that the validity of a reason would become even more important when the personal consequences are high. This was untested in Folkes' study.

In addition to personal relevance and distraction, the other variables discussed under Postulate 4 should also be moderators of the route to persuasion. For example, we have already noted that argument quality becomes a more important determinant of persuasion as people feel more personal responsibility for message evaluation (Petty *et al.*, 1980), and for individuals high rather than low in need for cognition (Cacioppo *et al.*, 1983). Although it has not yet been tested, the ELM expects that peripheral cues in the persuasion context should generally be more important for group than individually responsible message evaluators, and for individuals low rather than high in need for cognition. Before concluding this section, we note two additional variables that appear to moderate the route to persuasion.

One previously unmentioned variable that appears to affect the extent of issue-relevant thinking is the modality of message presentation. In general, audio and video presentations compared to print give people less opportunity to process issue-relevant arguments because exposure is forced rather than self-paced. Thus, presenting messages in written form should be especially important when the arguments are complex and difficult to process rapidly (Chaiken & Eagly, 1976). On the other hand, if it is generally more difficult to process issue-relevant arguments when exposure is forced rather than self-paced, simple cues in the persuasion context should be more powerful determinants of persuasion in the former than in the latter modality. Studies which have manipulated medium of presentation and source cues have supported this proposition. Thus, both source credibility (Andreoli & Worchel, 1978) and likability (Chaiken & Eagly, 1983) have had a greater impact on attitudes when a message was presented on video or audio tape rather than in written form.

Interestingly, the nature of the message itself has also been implicated as a determinant of whether a person processes mostly issue-relevant arguments, or searches for simple cues to determine message acceptability. For example, research suggests that messages that are either overly vague (Pallak *et al.*, 1983), or overly quantified (Yalch & Elmore-Yalch, 1984), may induce reliance on peripheral cues. The ELM would expect this to occur to the extent that these messages reduce either subjects' ability (vague message) or motivation (overly quantified message) to process issue-relevant arguments (Witt, 1976).

VII. Postulate 6: Biased Elaboration

We have now seen that a wide variety of variables can moderate the route to persuasion by increasing or decreasing the extent to which a person is motivated or able to process the issue-relevant arguments in a relatively objective manner. As we noted in discussing Postulate 3, however, variables can also affect persua-

sion by affecting motivation and/or ability to process message arguments in a more biased fashion. Specifically, Postulate 6 states:

Variables affecting message processing in a relatively biased manner can produce either a positive (favorable) or negative (unfavorable) motivational and/or ability bias to the issue-relevant thoughts attempted.

As we will see, there are a number of ways to induce biased processing, but often the bias results from a person's initial attitude becoming a more important schema in guiding processing (e.g., Tesser, 1978). Panel IV in Fig. 2 graphs the expected results for a variable that biases information processing activity. In the left half of the panel, the effects of a variable that produces a positive cognitive bias (enhancing favorable thoughts and/or reducing negative thoughts) is depicted. It is instructive to compare this pattern with the pattern of data in the two panels above it. First note that *unlike* a variable operating as a simple positive cue (left half of panel II), a variable producing a positive processing bias is not expected to affect all messages equally. Since the pure cue processor is not elaborating message arguments at all, the effectiveness of the cue is not constrained by the arguments presented. The biased processor, however, is attempting to process the arguments and in this regard is similar to the objective processor. Nevertheless, an important difference between objective and biased processing exists. The objective processor is motivated or is able to discover the "true validity" of the message, and thus strong arguments induce more persuasion and weak arguments induce less persuasion with more processing. In stark contrast, the biased processor is either particularly motivated or able to generate a particular kind of thought, often in defense of an initial attitude. However, even though the person is biased in processing a communication, the arguments in the message pose some limitation on this bias. For example, consider a person who is truly motivated to *counterargue* (and not simply discount) an advocacy. This person's task is simpler to the extent that the message provides weak rather than strong arguments in support of its position (see right half of Panel IV, Fig. 2).

Figure 2 summarizes the ways in which a treatment can affect attitude change according to the ELM, and it shows how these different processes can be tested by varying argument quality. First, a treatment can have no effect on persuasion for either strong or weak arguments (such as a peripheral cue under conditions of high elaboration likelihood; Panel I). Second, a treatment may produce only a main effect (Panel II). If so, it suggests that the treatment is operating as a simple positive or negative cue (low elaboration likelihood conditions). However, if a treatment interacts with message quality, it suggests that the treatment is affecting the elaboration likelihood. If the interaction follows the form depicted in Panel III of Fig. 2, it suggests that the processing is relatively objective. If a treatment main effect and an interaction as depicted in Panel IV of Fig. 2 is obtained, it suggests that the treatment is biasing information processing.

weak message that supported their own opinions on an issue (e.g., against government support of abortion), and the message either employed a religious or a legalistic perspective in the arguments presented. An analysis of subjects' ratings of message persuasiveness revealed a schema type $\times$ message type interaction: the legalistic message was seen as more persuasive by the legalistic than the religious subjects, and the religious message was seen as more persuasive by the religious than the legalistic subjects. In addition, recipients generated more topic-relevant thoughts when the message was reflective than when it was unreflective of their self-schema. Further analyses revealed that this effect was accounted for mostly by the increased generation of favorable thoughts to schema-reflective messages.

If a message is inconsistent with a person's initial opinion, however, it would be expected that prior knowledge would enhance the person's ability to counterargue the message. In a test of this hypothesis, Wood (1982; Experiment 1) assessed the prior knowledge and experience people had on the issue of environmental preservation by asking them to list their beliefs and previous behaviors concerning environmental preservation. Subjects were divided into high and low belief and behavior retrieval groups based on a median split on the number of beliefs and behaviors listed. Consistent with the view that this assessment technique taps prior knowledge, subjects who generated more behaviors indicated that they had thought more about preservation, knew more about the topic, and were more involved than subjects who generated fewer behaviors (no effects were found for belief retrieval, however). One to two weeks later, subjects returned and read a counterattitudinal message providing four arguments against environmental preservation. After message exposure, subjects reported their attitudes and gave their thoughts. Subjects who had high prior knowledge changed less in the direction of the message than subjects with low prior knowledge. In addition, subjects with high prior knowledge (as assessed by behavior retrieval) generated more counterarguments and fewer favorable thoughts in response to the message. In sum, the available research is generally consistent with the view that prior knowledge enables counterarguing of incongruent messages (Wood, 1982) and bolstering of proattitudinal ones (Cacioppo *et al.*, 1982).

2. Cue Effects

Research is also generally consistent with the view that simple cues or decision rules are more likely to affect susceptibility to influence when prior knowledge is low rather than high. One cue that has been studied in the context of previous issue-relevant knowledge is gender. Previous studies of sex differences in persuasion have provided some support for the view that females are more susceptible to influence than males in some contexts (see reviews by

Cooper, 1979; Eagly & Carli, 1983), and one explanation for this effect is based on the idea that females may have been socialized to be more agreeable (i.e., concerned with social harmony) than males (e.g., Eagly, 1978). To the extent that females have learned to be more agreeable and less dominant than males, the invocation of this socialized female gender role or category (cf. Deaux, 1984) could lead to a sex difference in influenceability. However, according to the ELM, attitude expression based on the female gender role should be more likely when women have little ability to process the issue-relevant information presented than when ability is high.

In a test of this hypothesis, we exposed male and female undergraduates to photographs relevant to domains for which men and women had rated their knowledge differently (Cacioppo & Petty, 1980b). Half of the photos depicted football tackles (high male knowledge) and half depicted current fashions (high female knowledge). Each photograph was accompanied by a set of comments attributed to another subject. The comments were either completely factual and descriptive (e.g., the dress is blue, the runner's feet are off the ground) or included an evaluation (e.g., that's a great tackle) that was either accurate or inaccurate. Subjects were asked to rate the extent to which they agreed with the comments made by the other subject. The ELM suggests that to the extent that gender roles provide simple rules as to how one should behave (e.g., "As a woman, I should maintain harmony"), such rules should operate mostly when ability (and/or motivation) to evaluate the stimuli are low. In our study, when the comments were completely factual and easily verifiable, both males and females should be equally able to evaluate the comments whether they concerned fashions or football; thus, there should be no sex differences in extent of agreement. When the comments were evaluative rather than descriptive, however, knowledge is required to confidently evaluate the statements. For football tackles, then, the invocation of the female gender role should lead to women showing more agreement than men whether the evaluations were accurate or inaccurate. Actual accuracy should make little difference because in both cases women would have little confidence (due to low knowledge) in their judgments. When the judgments concerned fashions, however, women do have the requisite knowledge and confidence to make judgments. Thus, they should be more accepting of the accurate evaluations, but less accepting of the inaccurate evaluations than men. The data from our study generally conformed to this pattern. Other research has also supported the view that prior knowledge is an important determinant of sex differences in influenceability (e.g., Karabenick, 1983; Sistrunk & McDavid, 1971).

Another simple rule that people sometimes use is based on observation of their own behavior and the situational constraints imposed upon it (i.e., the self-perception principle; Bem, 1967, 1972). For example, if an initially agreeable behavior is overjustified, people may reason that their behavior is governed more

B. FOREWARNINGS

Just as some variables generally enhance a person's ability to engage in biased processing of a persuasive message, such as prior knowledge, so too can variables enhance a person's *motivation* to process a message in a biased fashion, even if ability is held constant. For example, McGuire (1964) argued that inoculation treatments enhance resistance to persuasion mostly by increasing peoples' motivation to defend their beliefs. The persuasion literature has identified other variables that increase motivation to defend beliefs, and the most researched category of variables is "forewarning" (see Smith, 1982).

Papageorgis (1968) noted that two conceptually distinct kinds of warnings have been studied by persuasion researchers. One kind of treatment forewarns message recipients of the upcoming topic and/or position of the persuasive message (warning of message content). A second kind of treatment suggests to subjects that they are the targets of an influence attempt (forewarning of persuasive intent). Although some studies have explored the effects of combining the two kinds of forewarnings (e.g., Allyn & Festinger, 1961; Brock, 1967), it is possible to study their effects separately. Also, although many studies have shown that forewarnings can reduce the persuasive impact of a message, other studies have shown that forewarnings can enhance persuasion (e.g., Cooper & Jones, 1970; Mills, 1966). Importantly, which effect is obtained appears to depend largely on a person's motivation and ability to think about the issue. With low motivation and/or ability, forewarnings have tended to either have no effect or to enhance change in the direction of the advocacy. With high motivation and ability, however, resistance to persuasion has generally resulted (e.g., Apsler & Sears, 1968; Freedman, Sears, & O'Connor, 1964; Petty & Cacioppo, 1979a). According to the ELM, when motivation (e.g., personal relevance) and/or ability (e.g., prior knowledge) to think about an issue are low, forewarnings should enhance the salience of various cues (e.g., attractive sources; e.g., Mills & Aronson, 1965) or motives (e.g., impression management; e.g., Cialdini, Levy, Herman, & Evenbeck, 1973) in the situation that are capable of producing attitude change without issue-relevant thinking. When motivation and ability are high, however, forewarnings should modify attitudes by affecting issue-relevant thinking.

1. Warning of Message Content

A forewarning of message content gives message recipients advance indication of what the message is about. McGuire and Papageorgis (1962) hypothesized that the advance warning would motivate recipients to begin considering information that would support their beliefs and counterargue opposing arguments (i.e., biased processing). Consistent with this view, a content forewarning

is most effective when there is a sufficient time delay between the warning and message to allow thinking (Freedman & Sears, 1965; Hass & Grady, 1975; Petty & Cacioppo, 1977). In addition, studies employing the thought-listing procedure (e.g., Brock, 1967; Petty & Cacioppo, 1977) and psychophysiological assessments (Cacioppo & Petty, 1979a) have supported the view that when confronted with an impending counterattitudinal message on an involving issue, people use the period between the forewarning and the message to bolster their initial opinions. This "biased scanning" of arguments on the issue (cf. Janis & Gilmore, 1965) enables greater resistance to the subsequent message.

If accessing one's issue-relevant information prior to a persuasive attack assists in resisting the subsequent message, then a forewarning of the impending

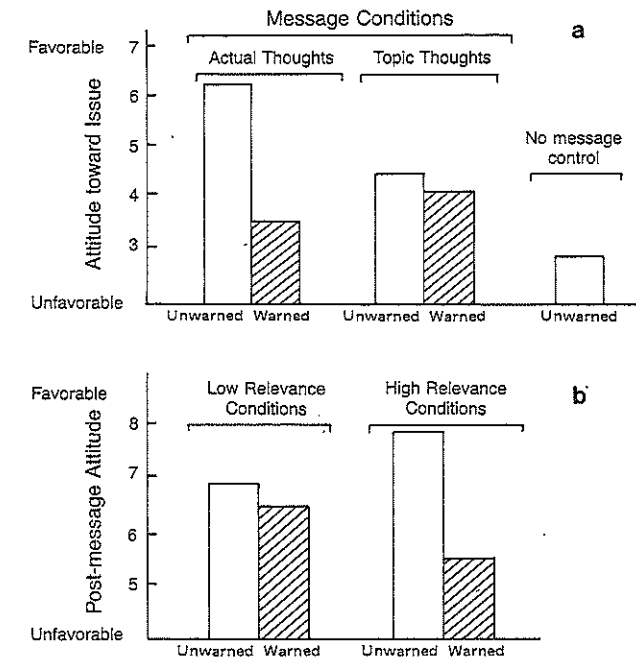


Fig. 7. Forewarnings bias message processing. (a) Forewarning of an involving counterattitudinal message topic and anticipatory topic-relevant thinking produce resistance to a message with strong arguments (data from Petty & Cacioppo, 1977; Experiment 2). (b) Forewarning of persuasive intent produces resistance to a message with strong arguments on a counterattitudinal topic when the issue is of high relevance (data from Petty & Cacioppo, 1979a).

relevance interaction on the postmessage attitude measure (see Fig. 7b). Although the warning decreased agreement overall, the effect was only significant under the high relevance conditions. The fact that the warning worked better under high than low relevance again suggests that the warning is not operating as a simple rejection cue. As we detailed in Section VI,A, cues tend to work better under low than high relevance conditions. Also, since the warning reduced persuasion even though the arguments were strong, this suggests that the warning induced biased rather than objective processing. When subjects were not warned, increasing involvement enhanced persuasion as would be expected if the arguments were strong and relevance increased subjects' motivation to process the arguments in a relatively objective manner (Petty & Cacioppo, 1979b). It appears that when a forewarning of persuasive intent was introduced under high involvement, the nature of the information processing changed as subjects became less objective, and more intent on finding fault with the message arguments in order to reassert their attitudinal freedom. Consistent with this reasoning, under high involvement, subjects who were warned generated significantly more counterarguments and fewer favorable thoughts than unwarned subjects in a postmessage thought listing.

C. OTHER BIASING TREATMENTS

Although we have focused on two highly researched variables that appear to enable or motivate biased information processing (prior knowledge and forewarning), other treatments may also bias the nature of message processing. We have already noted that McGuire's (1964) discussion of inoculation treatments provides a cogent example.

In our own research we have suggested several procedures for biasing the processing of a persuasive message. For example, in one study (Petty & Brock, 1979) we embedded a bogus personality assessment within an overall Barnum personality description (e.g., Forer, 1949). Subjects who in one experiment were led to believe that they had "closed-minded" personalities, subsequently generated a more one-sided profile of thoughts than subjects who were led to believe that they were "open-minded." In another study (Wells & Petty, 1980), we attempted to bias thought production by instructing subjects to make vertical or horizontal head movements while they processed a persuasive message. Whether the message was pro or counterattitudinal, subjects who engaged in vertical (yes) movements agreed with the message more than subjects who engaged in horizontal head movements. In other studies we have found that excessive message repetition (Cacioppo & Petty, 1979b) and the presence of hecklers (Petty & Brock, 1976) led to reduced agreement even though the arguments presented were strong.

The research on each of these independent variables (e.g., heckling, excessive repetition) clearly indicates that they are unlikely to enhance objective information processing. However, although there is some correlational evidence (e.g., from thought listings) to suggest that some of these treatments bias the nature of information processing, the studies may also be interpreted as indicating that the treatments serve as simple acceptance or rejection cues. For example, horizontal head movements, hecklers, or excessive repetition may not facilitate the production of counterarguments, but may instead induce negative affect that becomes associated with the advocacy (see Zajonc & Markus, 1982). Alternatively, the negative affect may bias information processing by increasing access to other information linked to negative states (Bower, 1981; Clark & Isen, 1982). Research that includes these treatments along with message quality and other motivational and ability variables should allow more definitive distinction of these possibilities (see also Section IX,B).

VIII. Postulate 7: Consequences of Elaboration

In the preceding sections of this article we have outlined how the ELM accounts for the initial attitude changes induced by persuasive messages. Postulate 7 deals with the different consequences of attitude changes induced via the central and the peripheral routes. Specifically:

Attitude changes that result mostly from processing issue-relevant arguments (central route) will show greater temporal persistence, greater prediction of behavior, and greater resistance to counterpersuasion than attitude changes that result mostly from peripheral cues.

There are several reasons why these differential consequences would be expected. Recall that under the central route, attitude changes are based on a thoughtful consideration of issue-relevant information and an integration of that information into an overall position. Under the peripheral route, however, an attitude is based on a simple cue that provides some affective association or allows some relatively simple inference as to the acceptability of the message. Thus, attitude changes induced via the central route involve considerably more cognitive work than attitude changes induced under the peripheral route. The process of elaborating issue-relevant arguments involves accessing the schema for the attitude object in order to evaluate each new argument (e.g., by comparing it to information previously stored in memory). Under the peripheral route, however, the schema may be accessed only once to incorporate the affect or inference elicited by a salient cue. Or, a peripheral schema unrelated to the issue

schema

this message contained six strong arguments in support of senior comprehensive exams. The other half of the students received a message from a low prestige, inexpert source; this message contained six weak arguments in support of senior comprehensive exams.

Based on our previous research (Petty & Cacioppo, 1979b; Petty, Cacioppo, & Goldman, 1981; see Section VI.A), and pilot testing of the sources and messages under high and low relevance conditions, we expected both the high and low involvement groups who received the positive source-strong arguments message to show equivalent amounts of initial persuasion. However, the change in the high relevance group should be based mostly on a careful evaluation and elaboration of the strong issue-relevant arguments, whereas the change in the low relevance group should be based mostly on the positive source cue. Similarly, the rejection of the advocacy in the high and low relevance groups exposed to the negative source-weak arguments message should be equivalent initially, but in the high involvement group the rejection should be based mostly on scrutiny of the weak arguments, whereas in the low relevance group the rejection should be based mostly on the negative source.

An analysis of subjects' immediate postmessage attitudes concerning senior comprehensive exams provided support. Both high and low relevance groups of subjects exposed to the strong message/source were more favorable than controls, and both groups of subjects exposed to the weak message/source were less favorable than controls. More interestingly, however, the degree of personal relevance had an impact on whether or not these initial attitudes persisted. From 10 to 14 days following message exposure, subjects were called by phone and were asked their opinions concerning a number of campus issues including the general idea of senior comprehensive exams. An analysis of variance (ANOVA) on the initial and delayed attitudes of high relevance subjects revealed only a main effect for type of communication. The positive source-strong arguments message was more effective than the negative source-weak arguments message both initially and at the delayed testing. An analysis on the attitudes of low relevance subjects, however, revealed a communication $\times$ time of measurement interaction. For these subjects, the initial difference between the two message conditions was no longer apparent at the delayed testing. In short, those subjects who formed their initial attitudes based on a careful consideration of issue-relevant arguments (high relevance) showed greater persistence of attitude change than those subjects whose initial attitudes were based primarily on the source cue (low relevance).

Other persuasion studies also support the view that conditions that foster people's ability or motivation to engage in issue-relevant cognitive activity enhance the persistence of persuasion. Thus, using more interesting or involving issues about which subjects have more knowledge (e.g., Ronis, Baumgardner, Leippe, Cacioppo, & Greenwald, 1977), providing more time to think about the

message (Mitnick & McGinnies, 1958), leading people to believe that they will be interviewed on the attitude issue (Chaiken, 1980), increasing message repetition (Johnson & Watkins, 1971), and reducing distraction (Watts & Holt, 1979) have all been associated with increased temporal persistence of attitude change (see Cook & Flay, 1978; Petty, 1977; for reviews, and Petty & Cacioppo, 1986, for a discussion of how the ELM differs from alternative models of attitude persistence, such as Kelman, 1961).

B. ATTITUDE-BEHAVIOR LINK

The previous section provided support for the view that attitude changes based primarily on thoughtful consideration (or self-generation) of issue-relevant arguments produced more enduring persuasion than changes based primarily on simple cues in the persuasion context. Research is also consistent with the view that attitude changes induced via the central route are more predictive of behavior than changes induced via the peripheral route.

As we noted earlier, perhaps the most effortful form of processing occurs when attitude change results from the self-generation of arguments. These changes, then, should be especially predictive of behavior. In a relevant program of research, Fazio and Zanna (1981) explored the consequences of attitudes formed via direct rather than indirect experience. When an attitude is formed via direct personal experience, the attitude is necessarily based on self-generated information. When an attitude is based on indirect experience (i.e., a message from others), less effortful processing may be involved. In some sense then, the distinction between direct and indirect experience is analogous to the distinction between attitudes based on role-playing (i.e., self-generation of arguments) versus passive exposure. Importantly, the research on direct versus indirect experience clearly indicates that the former attitudes are better predictors of behavior than the latter (see Fazio, 1985). The ELM suggests that one reason for this is that attitude formation based on direct experience may typically require more effortful elaboration of the merits of the object (e.g., puzzle; Regan & Fazio, 1977) than attitude formation based on passive exposure.

In our own research, we have also found that conditions that foster a high elaboration likelihood produce higher attitude-behavior correlations than conditions in which the elaboration likelihood is low. For example, in one study (described previously in Section VI.A.1), we exposed subjects to mock magazine advertisements for a disposable razor under conditions of either high or low personal relevance (Petty, Cacioppo, & Schumann, 1983). The ads that subjects saw contained either strong or weak arguments for the product and featured either a famous (likable) or an ordinary endorser. In addition to assessing product attitudes in this study, we also asked subjects to rate how likely they were to

ment that enhances or reduces the person's motivation and/or ability to counterargue a subsequent opposing communication. This work is consistent with the ELM in that it demonstrates that attitudes can be made more resistant by motivating or enabling people to engage in additional thought about the reasons or arguments supporting their attitudes. To date, however, no research has explicitly tested the ELM prediction that the manner in which an attitude is formed or changed has important implications for the resistance of the attitude. Specifically, the ELM predicts that people who come to accept an issue position because of a peripheral cue (e.g., source expertise) should be more susceptible to an attacking message than people who adopt the same issue position based on a careful scrutiny and elaboration of the message arguments.

IX. Complicating Factors

We have now presented the major postulates of the Elaboration Likelihood Model and some research relevant to these postulates. In reviewing the evidence for the ELM we have focused deliberately on variables and instances that were straightforward and relatively unambiguous in interpretation. Although it would be nice if we could provide an exhaustive list of variables that serve as peripheral cues and variables that affect message processing in either an objective or a biased manner, we have already seen that this is not possible. For example, we have argued that the effects on information processing of some variables may shift from relatively objective to relatively biased as the variable reaches very high levels. For example, although increasing personal relevance and message repetition may generally enhance subjects' motivation and/or ability to see the merits of strong arguments and the flaws in weak ones, we have suggested that when personal relevance or message repetition reaches very high levels, the initially objective processing may become biased as the person becomes motivated to reject the advocacy (Cacioppo & Petty, 1979b; Petty & Cacioppo, 1979b). In short, some variables have multiple effects on information processing. In addition, we have seen that some variables may affect information processing under certain conditions, but serve as peripheral cues in other contexts. For example, we reviewed evidence in Section VI,A that manipulation of the number of arguments in a message could serve as a peripheral cue when the personal relevance of the message was low, but that increasing the number of arguments in a message could increase the amount of information processing activity when the personal relevance of the message was high (Petty & Cacioppo, 1984a). In this section we will comment briefly on these and other intricacies of the ELM.

A. VARIABLES WITH MULTIPLE EFFECTS ON ELABORATION

In most of the research that we have discussed so far, we have examined the isolated effects of different source, message, recipient, and channel factors on information processing. However, in most natural persuasion situations, many variables combine to create the overall persuasion context. For example, consider a high need for cognition person who is part of a jury whose members share responsibility for evaluating an expert witness who presents weak arguments in a corporate tax case in a courtroom with noisy distractions. All of the many variables present in this situation must be considered jointly to determine the probable persuasive impact of the testimony. Normally, sharing cognitive responsibility with a group reduces information processing activity (Petty, Cacioppo, & Harkins, 1983), but our message recipient dispositionally tends to like to think (Cacioppo, Petty, & Morris, 1983) and is therefore less susceptible to motivation loss in groups (Petty, Cacioppo, & Kasmer, 1985). Therefore, *motivation* to process the message is likely to be high despite the group responsibility. However, due to a lack of prior knowledge about corporate taxes and the distractions inherent in the situation, our message recipient may have little *ability* to process the weak message arguments (Petty, Wells, & Brock, 1976; Wood, Kallgren, & Priesler, 1985). Thus, the perceived expertise of the witness may serve as a potent influence cue (Kiesler & Mathog, 1968).

Our example assumes that each of the features of the persuasion situation (e.g., distraction, group responsibility) can be considered separately and independently regardless of the levels of the other variables with which it is combined. If so, one can roughly add (subtract) the effects of each variable to determine the overall elaboration likelihood. Although this is often possible, as we discuss next, it is also possible for one variable to have very different effects on information processing depending on the level of other variables.

For example, some variables may increase information processing at one level of another factor, but may actually decrease processing at a different level of that factor. In one study exploring this possibility, we varied the personal relevance of a message, whether concluding summaries of the message arguments were framed as statements or as rhetorical questions (cf. Zillmann, 1972), and whether the arguments presented were strong or weak. In this study (Petty, Cacioppo, & Heesacker, 1981), all subjects heard over headphones a message advocating that seniors take a comprehensive exam in their major as a requirement for graduation. The study was designed to test our view that summarizing arguments as rhetorical questions (e.g., "Wouldn't instituting a comprehensive exam be an aid to those who seek admission to graduate and professional schools?") rather than as declarative statements, would motivate more thinking about the arguments. If rhetorical enhance relatively objective processing, then

B. VARIABLES THAT AFFECT ELABORATION AND SERVE AS CUES

In Section IX,A we argued that whether or not a particular variable enhances or diminishes processing, or motivates relatively objective or relatively biased processing, may depend on the level of other variables in the persuasion context. Similarly, whether a variable affects information processing or serves as a peripheral cue may depend on the level of other elements in the persuasion situation. We discuss this feature of the ELM below.

1. *Source Expertise/Attractiveness*

In Section I, we noted that one aspect of persuasion research that has disappointed reviewers of the field is that even variables that were expected to be quite simple in their effects on attitude change have instead proved to be quite complex. We also noted that perhaps the most dramatic example of this was the conflicting results of research on features of the message source (Eagly & Himmelfarb, 1974). Postulate 3 (Section IV) of the ELM outlines the several different ways in which source (and other) factors can affect persuasion: they can serve as arguments, they can serve as cues, or they can affect argument processing. In the research that we have reviewed so far, we have focused on how source factors operate when the elaboration likelihood is either very high or very low. We have seen that when people are unmotivated and/or unable to process a message, they rely on simple cues in the persuasion context, such as the expertise or attractiveness of the message source, although other cues may be used if they are more salient. Importantly, since subjects are either unmotivated or unable to evaluate message arguments, a positive source tends to enhance persuasion and a negative source tends to reduce persuasion, regardless of message quality (e.g., see top panels in Fig. 4).

On the other hand, when people are highly motivated and able to process message arguments, strong arguments are more effective than weak ones despite the presence of peripheral cues such as source credibility and attractiveness (e.g., see bottom panels in Fig. 4). When motivation and ability to process are high, people are concerned with evaluating the true merits of the advocacy. In order to do this, they will scrutinize all available and inferred information in the immediate persuasion context, and attempt to relate it to information stored previously in memory. Interestingly, a consideration of source factors may be part of a person's attempt to evaluate issue-relevant information when the elaboration likelihood is high. For example, under some circumstances a source feature may itself serve as a persuasive argument by providing information central to the merits of the attitude object (e.g., a physically attractive source, without saying anything, may provide persuasive visual testimony as to the effectiveness of a

beauty product; Kahle & Homer, 1985; Petty & Cacioppo, 1981b). Additionally, a consideration of source information might help a person in evaluating the true merits of any given argument (e.g., is the expert source biased or does the source have a vested interest?).

It may now appear that the cases of high and low elaboration likelihood are quite clear. Source factors affect agreement with a message by serving as simple acceptance or rejection cues when the elaboration likelihood is low, but do not serve as simple cues when the elaboration likelihood is high. Instead, they are considered along with the message arguments in an attempt to evaluate the true merits of the advocacy. These conclusions, however, are only part of the story of how source factors impact on persuasion. As we noted in Section III,A, we view elaboration likelihood as a continuum anchored at one end by the peripheral route to persuasion, and at the other end by the central route. In all of our research described in the previous sections, we have attempted to create and describe relatively clear instances of central and peripheral routes to persuasion. Thus, for example, in our research on motivation to process, subjects were either highly involved with the topic (e.g., students were confronted with an advocacy that had implications for their own graduation; Petty, Cacioppo, & Goldman, 1981), or very uninvolved (e.g., the students were certain that there were no personal implications of the advocacy). The extreme high and low elaboration likelihood conditions have been quite useful for theory testing purposes and in explicating the two routes to persuasion. However, these conditions represent only part of the elaboration likelihood continuum.

Specifically, many day to day persuasion contexts are unlikely to be as high or as low in elaboration likelihood as the conditions we have deliberately created in our initial research. For example, people are sometimes uncertain as to the personal relevance of an issue, or have moderate rather than very high or very low knowledge on a topic. We have proposed that under more moderate conditions, people use source characteristics to determine how much to think about the message (Petty & Cacioppo, 1981a, 1984c). When the personal consequences or prior knowledge on an issue are moderate or unclear, people may not be sure if the message is worth thinking about or if they are able to do so. Under these circumstances, characteristics of the message source can help a person decide if the message warrants (or needs) careful scrutiny. In our own research on moderate levels of motivation to process, undergraduate students have been led to believe that a change in policy was being advocated for their university, but they were not told when or if this policy ever would be implemented. Thus, unlike our studies on high and low personal relevance (see Section VI,A), subjects could not be certain whether or not the change in policy would affect them.

In one study (Puckett, Petty, Cacioppo, & Fisher, 1983), for example, we told subjects that students in an evening undergraduate continuing education course had written essays on the issue of whether comprehensive exams should

The initial (conformity) explanation of multiple source effects is consistent with the view that the mere number of other people advocating a position serves as a simple peripheral cue as to the validity of the advocacy. The second (information processing) interpretation, however, is more consistent with the view that the attitude changes induced by multiple sources follow the central route to persuasion (Harkins & Petty, 1983). The ELM, of course, suggests that both of these processes may operate in different situations. When the elaboration likelihood is very low (such as when personal relevance is low or distraction is high), people will be unmotivated to evaluate the issue-relevant information presented and may use the number of people who support the issue as a simple cue as to the worth of the proposal. When the elaboration likelihood is moderate, people may use the number of sources advocating a position as an indication of whether the message is worth considering. Finally, when the elaboration likelihood is very high, message recipients will undertake a deliberate assessment of the message arguments and the number of endorsers will have little further value as a motivator of thought or as a simple acceptance cue. No experiment to date, however, has examined the impact of the number of sources across the full elaboration likelihood continuum.¹³

Factors associated with the message source, of course, are not the only variables that can both serve as cues and affect message processing. Message variables can likewise serve in both roles. We have already discussed how the number of arguments could serve as a simple cue when personal relevance was low, but affect information processing when personal relevance was high (Petty & Cacioppo, 1984a). Similarly, recipient and context variables may serve in multiple roles. For example, we have shown that the physical posture of a message recipient can affect the extent of elaboration under moderate involvement conditions. In one study (Petty, Wells, Heesacker, Brock, & Cacioppo, 1983), people who were reclining comfortably during message exposure showed greater attitudinal differentiation of strong from weak message arguments than people who were standing (see Box 6, Fig. 8). If subjects were presented with a message they were unmotivated or unable to elaborate, however, then posture (or other factors related to comfort during message exposure) might serve as simple positive or negative affective cues (e.g., Griffitt & Veitch, 1971).

Importantly, even though the ELM holds open the possibility that variables can affect agreement either by having an impact on information processing or by serving as simple cues, the ELM specifies, in a general manner at least, the conditions under which each process is likely to operate. Thus, a whole list of source (e.g., credibility, attractiveness, number of sources), message (e.g.,

¹³Our analysis of multiple sources assumes that all sources are advocating the same position. When conflicting positions are advocated by different numbers of people (as in minority influence), the situation becomes more complex (see Maass & Clark, 1984).

number of arguments, use of rhetorical, discrepancy), audience (e.g., recipient posture, presence of hecklers, false physiological feedback), and other variables may affect attitudes by modifying information processing under certain conditions (e.g., ambiguous personal relevance), but affect attitudes by serving as simple cues in other contexts (e.g., low prior knowledge).

X. Summary and Conclusions

At the most general level, we have outlined two basic routes to persuasion. One route is based on the thoughtful (though sometimes biased) consideration of arguments central to the issue, whereas the other is based on affective associations or simple inferences tied to peripheral cues in the persuasion context. When variables in the persuasion situation render the elaboration likelihood high, the first kind of persuasion occurs (central route). When variables in the persuasion situation render the elaboration likelihood low, the second kind of persuasion occurs (peripheral route). Importantly, there are different consequences of the two routes to persuasion. Attitude changes via the central route appear to be more persistent, resistant, and predictive of behavior than changes induced via the peripheral route.

In the body of this article we have discussed a wide variety of variables that proved instrumental in affecting the elaboration likelihood, and thus the route to persuasion. In fact, one of the basic postulates of the Elaboration Likelihood Model, that variables may affect persuasion by increasing or decreasing scrutiny of message arguments, was highly useful in accounting for the effects of a seemingly diverse list of variables (see Figs. 3 and 8). The effects of these variables had been explained with many different theoretical accounts in the accumulated persuasion literature. The ELM was successful in tying the effects of these variables to one underlying process. We have also seen that many different variables could serve as peripheral cues, affecting persuasion without issue-relevant thinking. Finally, we saw that some variables were capable of serving in multiple roles, enhancing or reducing thinking in some contexts, and serving as simple acceptance or rejection cues in others.

We began this article by noting that reviewers of the attitude change literature have been disappointed with the many conflicting effects observed, even for ostensibly simple variables. For example, manipulations of source expertise have sometimes increased persuasion, sometimes have had no effect, and have sometimes decreased persuasion. Similarly, studies testing different theories have sometimes found the theory to be useful in predicting attitude change, and at other times have found the theory to be unproductive. For example, self-perception processes appear to operate under some conditions, but not others.

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CONTRIBUTORS TO VOLUME 19

Robert S. Baron

John T. Cacioppo

Jeffrey D. Fisher

Dieter Frey

Arie Nadler

Richard E. Petty

David P. Phillips

Ivan D. Steiner

David A. Wilder

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Leonard Berkowitz

DEPARTMENT OF PSYCHOLOGY
UNIVERSITY OF WISCONSIN—MADISON
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